

Louis Messonnier

Atlanta, GA

404-797-8161 | louismessonnier@gmail.com

www.linkedin.com/in/louismessonnier | <https://github.com/louismessonnier>

EDUCATION

Georgia Institute of Technology, Atlanta, GA

May 2028

Bachelor of Science in Mechanical Engineering, Minor in Materials Science and Engineering

GPA: 4.00

EXPERIENCE

HyTech Racing Georgia Tech, Atlanta, GA

September 2024-Present

Powertrain Engineer

- Performed assembly, disassembly, and troubleshooting of powertrain components including gearbox, uprights, and cooling system
- Set up, repaired, and operated a cooling test setup to measure temperature, pressure, and flow rate using an Arduino, sensors, and data logging code
- Designed, prototyped, and finalized a catch can holder using CATIA and 3D printing and installed holder on the vehicle's cooling system
- Contributed to CAD design and integration of powertrain components in CATIA within Formula SAE constraints
- Evaluated component selection using datasheets and performance requirements to ensure compatibility within powertrain system

RESEARCH

Duke University Thomas Lord Department of Mechanical Engineering & Materials Science, Durham, NC

Summer 2025

Undergraduate Research Assistant

- Performed an extensive literature review on TPMS lattices, porous biomaterial mechanics, bone-implant interface behavior, and 3D-printed surface topography to support methodology and analysis
- Planned and carried out a self-guided study on the mechanical behavior of anisotropic gyroid unit cells under tensile and compressive loading
- Produced and tested 128 SLA-printed samples across two load types, two stretch directions, and four stretch ratios
- Processed and analyzed mechanical test data using MATLAB and Excel, including stress-strain computation, property calculation, and graphing
- Identified trends in anisotropic mechanical behavior across loading conditions, informing future lattice design considerations
- Built an Ansys tensile simulation of gyroid samples that qualitatively matched experimental fracture behavior, gaining experience in meshing, boundary conditions, and material modeling

SKILLS

Technical:

- **CAD:** SolidWorks, CATIA, nTop, Revit, KiCad
- **Simulation:** Ansys Mechanical
- **Programming:** MATLAB, Arduino
- **Fabrication:** SLA/FDM 3D printing, mechanical testing systems

Languages: English (native), Spanish (intermediate)

Relevant Coursework: Statics, Dynamics, Circuits, Electronics, Engineering Computing, Materials Science, Engineering Graphics

HONORS & INTERESTS

Honors: International Baccalaureate Diploma Recipient (May 2024), National Merit Semi-Finalist (2024), Eagle Scout Rank (December 2022)

Technical Interests:

- Automotive systems and powertrain design
- Additive manufacturing and lattice structures
- Mechanical testing and simulation